

CASR: Efficient Cascade Network Structure with Channel Aligned method for 4K Real-Time Single Image Super-Resolution

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Abstract

Recent advances in display technology have led to the widespread use of 1080p (FHD) and 4K (UHD) displays, increasing the importance of single-image super-resolution (SISR), a technique that uses low-resolution images to create high-resolution images. Existing research on SISR has focused on improving the accuracy in constructing deep learning networks with significant computational complexity. However, since devices such as TVs, VR headsets, and autonomous systems have limited memory, it is necessary to research real-time image super-resolution. In this paper, we propose CASR, an image super-resolution network that transforms compressed low-resolution images into 4K images in real-time. We propose a cascade upsampling with channel alignment method for upsampling, and reparameterization convolution layer to improve the performance of the network without any additional computational cost. In addition, we used the distillation method to improve performance and time. Experimental results show that the proposed network achieves 22.80 dB PSNR and 0.56 ms inference time on RTX 3090 devices when evaluated on the AIS 2024 real-time super-resolution validation scale X4 dataset. The code is available at <https://github.com/rlghksdbs/CASR>

1. Introduction

The SISR technique is a technique for generating high-resolution (HR) images from low-resolution (LR) images to enhance visual clarity and quality. SISR is one of the most widely used computer vision techniques in a variety of fields, including text restoration [48], underwater imaging [10], autonomous vehicle [46] [6], medical image analysis [41], satellite image analysis [40], remote sensing [43, 53], etc. Initially, interpolation methods [17, 45, 54]

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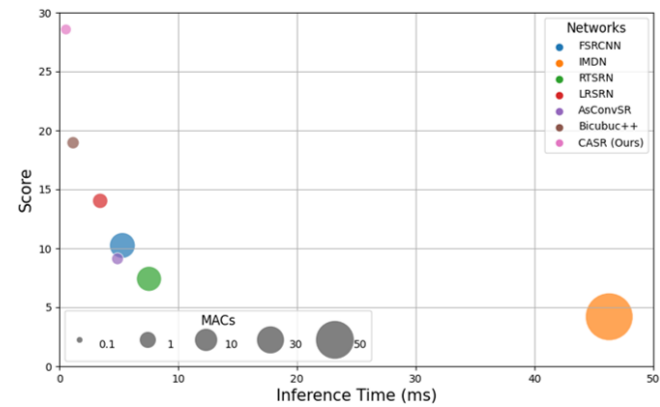


Figure 1. Comparison between the existing network and the proposed CASR for $\times 4$ upsampling score upsampling in terms of Score and Inference Time. Score is introduced in Sec. 3.1.

were used to solve this problem, and with the development of deep learning, deep learning-based SISR networks have been proposed [8, 9, 33].

Recent hardware advancements have led to the adoption of high-resolution displays in mobile phones, TVs, VR devices, and other devices, making the SISR technique more important. Therefore, research on SISR networks to handle high resolutions such as FHD (2K) and UHD (4K) has been actively conducted [11]. As a result, high-resolution SISR performance has improved, but the structure of most SR networks has become more complex and large and requires high computational capacity. This makes it difficult to run in real time on devices limited in resources [28]. Therefore, it is necessary to research lightweight and real-time SR networks.

To address the aforementioned issues, this paper proposed CASR, a real-time SR network that generates compressed 540p images into 4K high-resolution images. First, we review the factors that must be considered to build a lightweight network. Using this factor, we constructed

an efficient network and further applied reparameterization blocks and knowledge distillation techniques to improve performance while maintaining model size. Finally, we propose a cascade upsampling with channel alignment method for upsampling, which can improve performance and significantly reduce inference time. As a result, we obtained an inference time of 0.56 ms on RTX 3090 GPUs. Furthermore, as shown in Fig. 1, we can see that the score considering PSNR performance and inference time is superior to the state-of-the-art SISR task.

Our contributions can be summarized as follows:

- We propose a CASR network, an SR network with low computational complexity and extremely high speed that can perform in real-time on memory-constrained devices.
- We apply distillation techniques and propose an advanced reparameterized convolution layer that can improve performance while maintaining model size.
- We propose channel alignment with cascade upsampling scheme, an efficient upsampling scheme that can improve performance while reducing inference time. The results show improved performance and are faster than existing upsampling schemes.

2. Related Works

2.1. Lightweight Single Image Super Resolution

In recent years, various lightweight SISR networks have been proposed to run on devices with limited computing resources. Dong *et al.* proposed FSRCNN [16], which uses a deconvolution layer to solve the problem of the SISR method in which conventional networks are bicubic upsampled during preprocessing [15]. CARN [2] introduced lightweight SISR networks with a cascading structure in the model. Using a cascading model structure, CARN aimed to enhance the speed of SISR networks, enabling the CARN-M network to operate on actual mobile devices. To obtain distilled features, IDN [26] proposed an information distillation mechanism using channel splitting operation, which divides the feature into two parts in the channel dimension and retains one while passing the other through a subsequent convolution layer for refinement. IMDN [27] proposed the information multi-distillation block (IMDB), which repeatedly performs the channel split operation. RFDB [35] proposed a feature distillation connection (FDC) method using IMDN as the baseline and further used a shallow residual block (SRB) to improve performance and construct a lightweight model.

Moreover, recently, various challenges have been organized that aim to develop SR networks that can operate in real-time on GPUs and mobile devices. In the Mobile AI & AIM 2022 challenge [28], Gankhuyag *et al.* [18] proposed a

shallow SR network that can run on mobile NPUs by leveraging the reparameterization block. In the 2023 New Trends in Image Restoration and Enhancement (NTIRE) workshop, Guo *et al.* proposed Assembled Convolution [20] for real-time upscaling to 4K resolution on GPUs. Moreover, various techniques have been proposed, such as pruning to improve the inference time [4] or using reparameterization blocks even when the input and output channels are different [19].

2.2. Reparameterization

Recently, there has been a growing emphasis on the importance of the reparameterization method [13, 14, 19], which has been applied to various computer vision tasks such as object detection [47], architecture search [24], and pruning [31]. Reparameterization involves transforming different architectures into each other by equivalently adjusting the parameters. For instance, a 1×1 convolution branch and a 3×3 convolution branch can be converted into a single 3×3 convolution branch [14]. During the training phase, complex topologies with multiple branches [13, 14, 18, 19] were used to replace traditional linear layers to improve model performance. [7] have explored the merging of depth-wise separable convolution kernels during training. Subsequently, during inference, the refined models used during training are transformed into simpler ones to accelerate inference. [3] proposed collapsible reparameterization block that overcomes the problem of vanishing gradients if not properly augmented with residual connections of [14]. [18] applying the reparameterization block for a mobile NPU and adjusting the reparameterization block for the quantization step.

2.3. Knowledge Distillation

Knowledge distillation was first proposed by Hinton *et al.* [23], a method of transferring knowledge from the teacher model (large network) to the student model (small network) to improve the performance of the student model while reducing memory usage and computation. It was initially developed to compress models and accelerate inference on lightweight devices. Knowledge distillation has been used to efficiently construct small models and improve performance in a variety of fields, including natural language processing [29, 44], image classification [36, 50] and speech recognition [22].

The field of SISR also utilizes knowledge distillation to improve the performance of SISR models by using distillation in the image domain. Furthermore, several studies have proposed distillation methods for SR, such as feature distillation in the feature domain [21, 34, 39] to transfer features from a teacher model to a student model, and self-distillation [30], a form of self-supervision technique where the student model transfers knowledge to itself without a

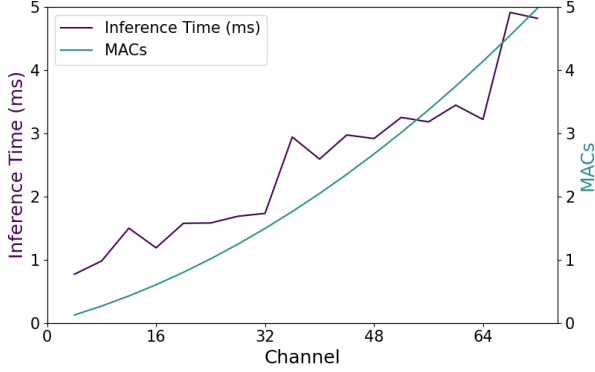


Figure 2. MACs and Inference Time based on convolution channel.

separate teacher network.

3. Proposed Methods

In this section, we first review the factors that need to be considered when designing a real-time network in Sec. 3.1. In Sec. 3.2, we describe the overall structure of the proposed network. Then we describe the reparameterization and distillation methods that we used in our network to improve the speed of the network. Finally, we describe our training strategy in Sec. 3.3.

3.1. Real-Time SISR Network Design

We aim to design real-time SISR networks to minimize the amount of computation and inference time while simultaneously achieving the highest performance. To effectively design a real-time SISR network, it is important to consider factors such as the number of convolution layers, the number of channels, and the connection between layers. The subsequent scores are utilized to determine the most efficient results, considering inference time, MACs (Multiply-Accumulate Operations) and PSNR (Peak Signal-to-Noise Ratio):

$$Score = \frac{2 \times 2^{(PSNR-thr)}}{0.1 \times \sqrt{InferenceTime}} \quad (1)$$

thr is 22.70 dB, the psnr of the model presented as a baseline in the AIS2024 Real-Time SR Challenge [51].

Channel: Depending on the number of channels in the convolution layer, the processing speed can be slowed for various reasons, such as memory bandwidth limitations on GPU devices or parallelism limitations. Therefore, it is important to set the appropriate number of channels to avoid speed degradation in the network, so we first explore the association between the inference time and the MACs according to the number of channels. As shown in Fig. 2,

the MACs increase linearly as the number of channels increases. However, for the inference time, we can observe a significant increase based on 32 and 64 channels. Hence, the network channels are restricted to a maximum of 32.

Backbone: In recent years, several real-time SISR networks have used subsequent convolutional layers rather than complex structures [4, 19, 20]. These networks have a head, which is a convolutional operation to convert the input image into feature space, a backbone to obtain deeper features, and a transition layer to align the channels for upsampling. To design a network, the number of layers in the backbone is also important. Tab. 1 shows that the highest number of backbones performs better, but considering runtime and MACs, it can be seen that a backbone with one convolution layer is suitable for real-time SR networks.

Method	PSNR	Runtime	MACs	Score
Backbone (4)	23.07	3.20	3.22	14.52
Backbone (3)	23.04	2.69	2.64	15.50
Backbone (2)	22.99	2.15	2.07	16.65
Backbone (1)	22.91	1.62	1.49	18.29

Table 1. Compare efficiency based on the number of backbones

Unshuffle: To calculate MACs for a deep learning network, the formula is as follows:

$$MACs = K^2 \times C_{in} \times C_{out} \times H \times W \quad (2)$$

Where K is the kernel size of the convolution, C_{in} is the input channel, C_{out} is the output channel, and H and W are the height and width of the input image, respectively. It is recommended to apply unshuffle to reduce H and W by downsizing the input image, which reduces the overall computation of the network. Recently proposed real-time SISR [11, 20] networks have adopted unshuffle for this reason. Based on Tab. 2, we downsize the input image by a factor of 3.

Method	PSNR	Runtime	MACs	Score
w/o Unshuffle	22.91	1.62	1.49	18.29
Unshuffle (2)	22.81	1.07	1.06	20.96
Unshuffle (3)	22.73	0.91	0.98	21.53
Unshuffle (4)	22.65	0.81	0.95	21.48

Table 2. Compare efficiency based on the unshuffle

3.2. Network Architecture

Based on Sec. 3.1, we propose the CASR network, a real-time SISR network that is effective in all aspects of

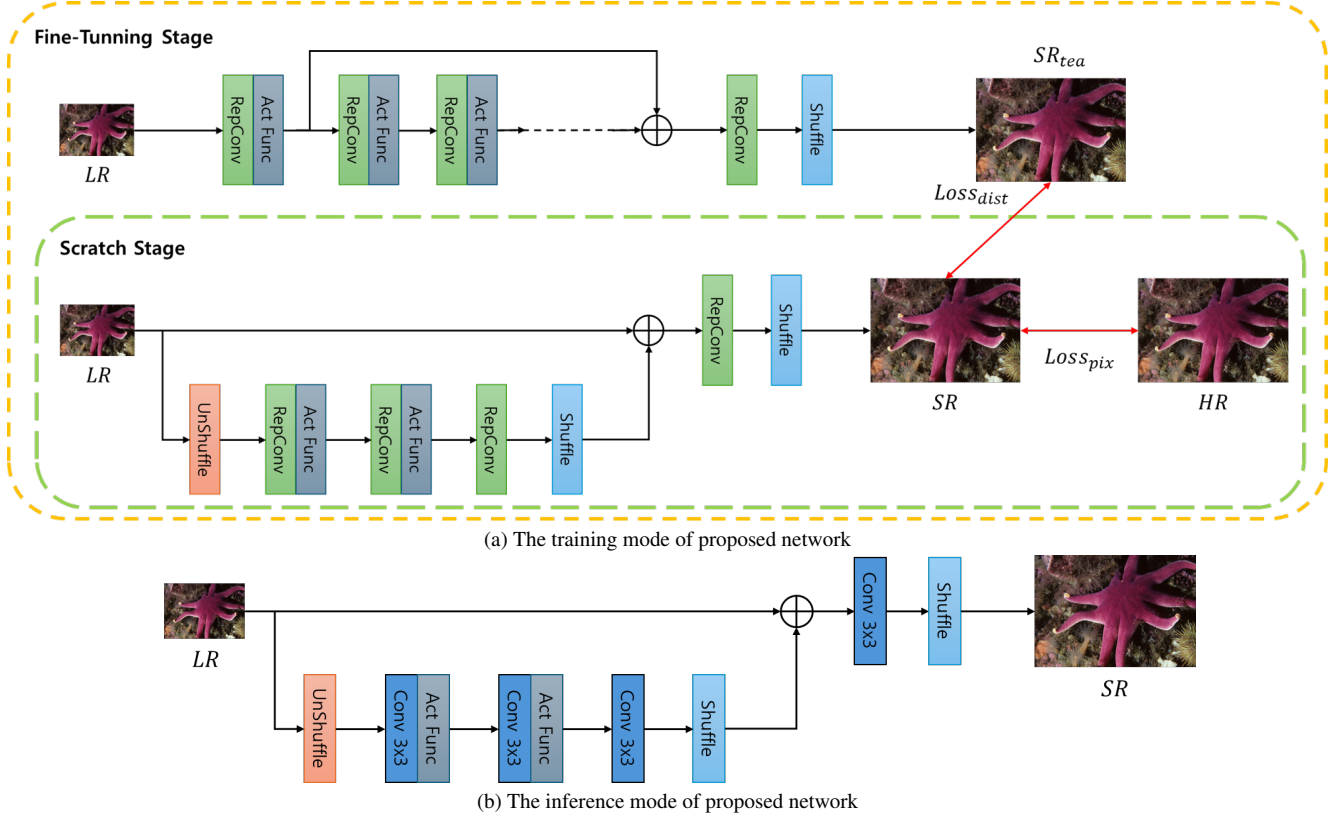


Figure 3. The overall structure of the CASR network, a lightweight real-time image super-resolution network.

PSNR, MACs, and runtime. The general structure of the CASR network is shown in Fig. 3b. We further utilize knowledge distillation and reparameterization techniques to improve the PSNR. And since we use a network structure that downsamples the input image by 3 times, a total of 12 times is required to create the HR image, which increases the inference time and degrades the performance, we propose the cascade with channel upsampling method to solve the problem.

Upsample with Channel Alignment: To upscale by $N\times$ using shuffle in SISR, the channel before the shuffle must be $N \times N \times 3$. In our proposed network, we perform $3\times$ downsampling, so we need to do $12\times$ to create a $4\times$ upsampled HR image. In this case, the total number of channels should be 432 for direct upsampling, as shown in Fig. 4a. This method has the drawback of increased computation due to the large channel difference between the input and output. When using the cascade method of upsampling $3\times$ and then $4\times$ as shown in Fig. 4b, the shuffle computation is reduced because the difference in channel between input and output is reduced. However, this method has the problem of increasing the runtime because the shuffle operation has to be performed twice.

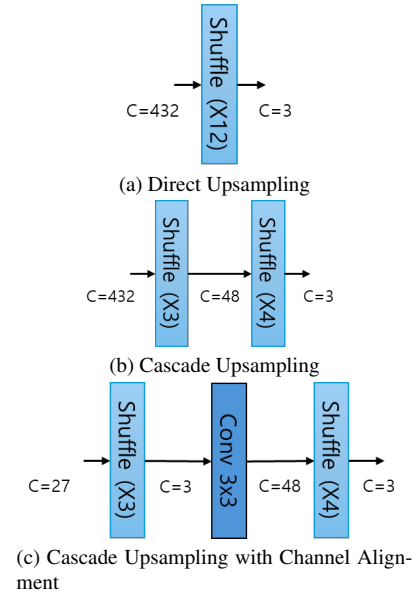


Figure 4. Upsampling Strategies

Therefore, we propose the cascade upsample with channel alignment as shown in Fig. 4c. To reduce the channel

gap, we first align the channels to 27 channels before shuffle for $3\times$ upsample. Then we use convolution to make it 48 channels for the $4\times$ upsample. As a result, as shown in Tab. 3, it can be seen that the runtime is reduced compared to the cascade method, although it uses additional convolution, and the amount of computation is also reduced compared to the direct method. Furthermore, the proposed method can utilize the information from the input image because it becomes 3 channels after upsampling by $3\times$, and when the input is added, the computation and runtime are maintained, and the performance increases.

Method	PSNR	Runtime	MACs	Score
Direct Upsample	22.73	0.910	0.980	21.53
Cascade Upsample	22.71	1.074	0.488	19.47
with Channel Alignment	22.71	0.725	0.253	23.68
with input	22.76	0.725	0.253	24.40

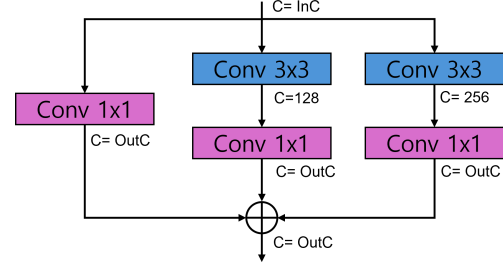
Table 3. Compare efficiency based on upsampling methods

Output Distillation: Our network has a lightweight structure, so it does not perform better than a larger network. Therefore, we use a knowledge distillation approach that uses a large network to feed the knowledge to CASR. The teacher network that we used was a simple structure with a deep Repconv layer, as shown in Fig. 3a. Finally, we only performed output distillation on $f^{tea}(LR)$, the result of the teacher model, and $f(LR)$, the result of CASD, using the following loss function:

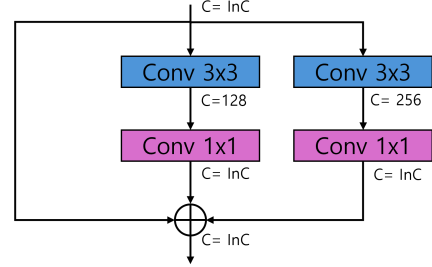
$$L_{dist}(\theta) = \frac{1}{n} \sum_{i=1}^n (f^{tea}(LR) - f(LR))^2 \quad (3)$$

Reparameterization Block: We utilized the reparameterization technique (RepConv) during the training phase to enhance the quality of the reconstructed images. An illustration of our proposed RepConv block is presented in Fig. 5. RepConv is capable of maintaining its linearity property, even when the convolution layer overlaps in different ways, thus enabling reconstruction. Previous work [7, 18] has applied RepConv only when the input and output channels of the convolution layer are equal. In contrast, we have extended the application of RepConv to all convolution layers, which proves to be more effective than restricting it to specific layers. Our RepConv module represents an enhanced module of the technique proposed in [19]. In contrast to [19], our RepConv module features three distinct branches, with the main two branches differing in the number of inner channel numbers 128 and 256. In cases where the input and output channels are equal ($InC = OutC$, InC denotes the input channel and $OutC$ denotes the output channel), we employ the RepConv block as shown in Fig. 5b. Where the input and output channels differ ($InC \neq OutC$), we utilize

an enhanced version of the RepConv block, as shown in Fig. 5a. This version incorporates a Conv 1×1 block instead of a skip connection. As a result, our RepConv block captures high-level information during training and can be fused to a Conv 3×3 layer during the inference stage.



(a) RepConv when in/out channels are not equal



(b) RepConv when in/out channels are equal

Figure 5. RepConv Block of training mode.

3.3. Training Strategy

We train our proposed model with three stages: scratch, fine-tuning, and lightweight-tuning stages.

3.3.1 Scratch Stage

In the scratch stage, we train our proposed model by cropping the HR and LR images into patches of sizes 384×384 and 96×96 , respectively. We use a batch size of 8 and the optimizer is Adam with a learning rate of 0.0005. We used a cosine warm-up scheduler and set a 0.1 percentage warm-up ratio. Finally, we use the $l1$ loss between the SR image and the HR image generated by the model as the loss function:

$$L_{l1}(\theta) = \frac{1}{n} \sum_{i=1}^n |f(LR^i) - HR^i|, \quad (4)$$

where θ denotes the trainable parameters of the proposed network, and n represents the quantity of training patch images. LR^i and HR^i refer to the images of the LR patch and their corresponding HR patch, respectively. The function $f(\cdot)$ defines the proposed network.

Scale	Network	Set5		Set14		B100		Urban100		Manga109		Average	
		PSNR/SSIM	Score	PSNR/SSIM	Score	PSNR/SSIM	Score	PSNR/SSIM	Score	PSNR/SSIM	Score	PSNR/SSIM	Score
X4	Bicubic	26.29/0.7503	-	23.52/0.6379	-	22.17/0.5533	-	21.29/0.6182	-	21.22/0.7008	-	22.90/0.6521	-
	FSRCNN [16]	21.19/0.7819	15.04	23.88/0.6545	7.15	21.98/0.5561	3.32	21.74/0.6474	8.12	21.24/0.7147	4.46	23.21/0.6709	7.62
	IMDN [27]	28.15/0.8151	19.24	24.28/0.6748	4.19	21.79/0.5595	0.86	22.32/0.6862	6.13	20.81/0.7332	0.83	23.47/0.6937	6.25
	RTSRN [18]	26.60/0.7552	5.59	23.66/0.6424	4.40	22.06/0.5496	3.10	21.45/0.6232	4.52	21.21/0.6897	3.60	23.00/0.6520	4.25
	LRSRN [19]	27.37/0.8007	24.11	24.07/0.6651	11.53	21.88/0.5584	3.60	22.04/0.6669	15.18	21.07/0.7273	4.35	23.29/0.6836	11.75
	AsConvSR [20]	26.43/0.7479	5.48	23.59/0.6382	4.96	22.07/0.5470	3.94	21.41/0.6214	5.31	21.24/0.6853	4.65	22.95/0.6479	4.87
	Bicubic++ [4]	26.57/0.7498	13.59	23.61/0.6393	10.46	22.00/0.5451	7.27	21.43/0.6208	11.23	21.19/0.6809	8.92	22.96/0.6471	10.29
	Proposed work	26.64/0.7579	21.62	23.70/0.6417	17.00	22.04/0.5500	11.13	21.57/0.6314	19.69	21.19/0.6884	12.76	23.03/0.6538	16.44

Table 4. Quantitative results comparison on benchmark datasets.

Scale	Network	DIV2K		AIS2024 val		MACs (K)	Inference Time (ms)
		PSNR	Score	PSNR/SSIM	Score		
X4	Bicubic	26.37/0.73	-	22.63/0.74	-	-	-
	FSRCNN	26.93/0.75	9.516	22.93/0.75	10.764	14.27	5.302
	IMDN	27.56/0.77	7.694	23.21/0.75	4.412	55.72	46.275
	RTSRN	26.59/0.74	4.945	22.727/0.74	7.785	13.35	7.548
	LRSRN	27.24/0.76	18.109	23.08/0.75	14.728	3.22	3.441
	AsConvSR	26.53/0.74	5.699	22.71/0.74	9.574	0.91	4.899
	Bicubic++	26.57/0.74	12.363	22.73/0.74	19.906	1.16	1.160
	Proposed work	26.68/0.74	20.675	22.80/0.75	29.986	0.25	0.564

Table 5. Quantitative results comparison on DIV2K and AIS2024 validation datasets.

3.3.2 Fine-Tuning Stage

In the second step, the model was initialized with the weights trained in the scratch stage. The output distillation method used at this stage. We use a fine-tuning loss l_{fine} , which improves the value of the peak signal-to-noise ratio (PSNR) by 0.01 $\sim$ 0.02 dB. In this step, the initial learning rate was set to 0.00005 and the Adamw optimizer was used along with a cosine warm-up. The total epoch was set at 800 epochs.

$$L_{l2}(\theta) = \frac{1}{n} \sum_{i=1}^n (f(LR^i) - HR^i)^2 \quad (5)$$

$$L_{fine}(\theta) = L_{l2}(\theta) + L_{dist}(\theta) \quad (6)$$

3.3.3 Lightweight-Tuning Stage

In this stage, we focused on reducing the inference time. The model weight was initialized using the weights trained in the previous stage. In addition, the distillation technique was applied in this phase as well. The training hyperparameters were kept identical to those in the second step. At this point, the bias term of the reparametrization block was deactivated, leading to a decrease in the inference time by 0.16 ms. Although there was a small decrease of 0.006 dB in the accuracy of the PSNR value, the overall score showed enhancement.

4. Experimental Results

4.1. Implementation Details

We use Pytorch version 1.13 for all training phases and use the DIV2K dataset [1] for the initial scratch training phase. In addition, we customized publicly available datasets to improve performance. This customized data set comprises the full DIV2K training set (800 images), the initial 1000 images from the Flickr [49] training set, 121 samples from the GTA [42] training sequences 00-19, and the first 1000 images from the LSDIR [32] dataset. The customized dataset is used for the fine-tuning stage and the lightweight-tuning stage. To generate compressed low-resolution images, we degrade random cropped images with avif compression with various compression factors. We use the real-time image super-resolution challenge in the AIS2024 script for a compressed low-resolution image [12]. Also, we use the AIS2024 val [12] dataset for evaluation. For both training stages, we used random cropping, rotation 90, and horizontal and vertical flip augmentation. The training process was executed using NVIDIA A6000 GPUs.

4.2. Compare With State-Of-The-Arts

In this section, we compare our proposed network with the state-of-the-art lightweight SR networks FSRCNN [16], IMDN [27], RTSRN [18], LRSRN [19], AsConvSR [20], and Bicubic++ [4], shown in Table. 4. The benchmarks used for the comparison are Set5 [5], Set14 [52], B100 [37], Urban100 [25], Manga109 [38]. We provide benchmarks to compare the performance of upsampling the compressed

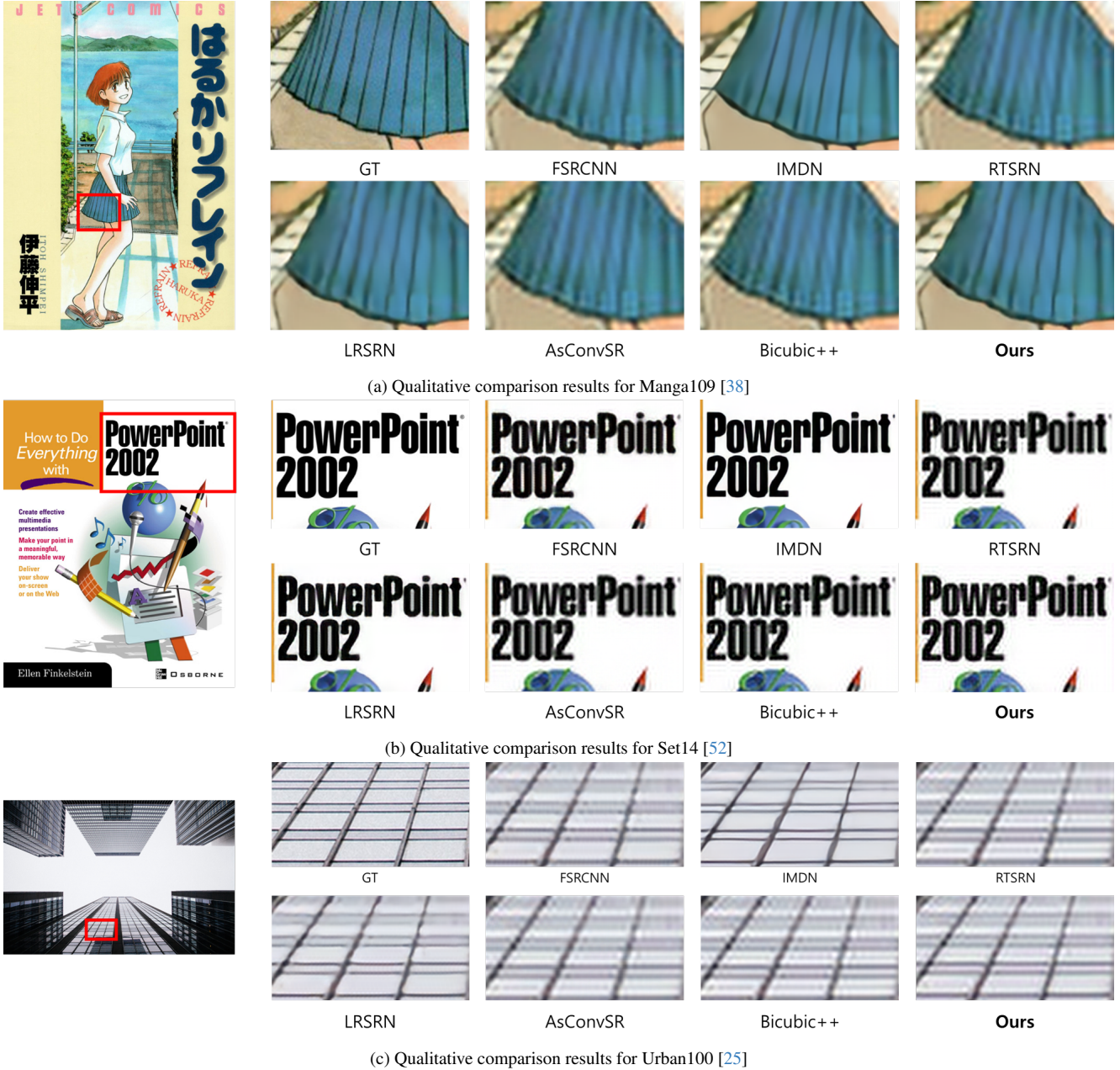


Figure 6. Qualitative comparison of proposed network with previous lightweight SR networks on scale $\times 4$.

image. We used PSNR and SSIM for comparison and used the scoring kernel described in Eq. (1), where the value of the scoring kernel thr is equal to the bicubic result. Furthermore, the inference time is measured on the GPU RTX3090, and the detailed inference time of each model is shown in Table 5. However, AsConvSR [20] and Bicubic++ [4] modified the version because the original design is designed for scale 2, so we changed the last number of convolution channels to a suitable scale 4. The average PSNR value of bicubic is 22.90 and the proposed work is 23.03, higher than

Bicubic by 0.13. Furthermore, our proposed work has a higher PSNR value to compare RTSRN [18] and AsConvSR [20]. However, the average PSNR value is lower than the FSRCNN [16], IMDN [27] and LRSRN [19] methods, but the average score value is higher than all previous methods. Moreover, we can see that our proposed network has a runtime of 0.564 ms and MACs of 0.25K, which is the fastest and lowest computation compared to the existing networks. This outcome demonstrates that our proposed approach is more effective than previous research within the scope of

Methods	AIS2024 Validation data			
	PSNR	Runtime (ms)	MACs (K)	Score
+ Baseline	22.9196	1.62	1.49	18.2969
+ Downsample with direct upsample	22.7389	0.91	0.98	21.5335 (+3.24)
+ Cascade upsample	22.728	1.05	0.48	19.8460 (-1.69)
+ Cascade upsample with Channel Alignment	22.755	0.72	0.253	24.4015 (+4.56)
+ RepConv	22.7844	0.72	0.253	24.8636 (+0.46)
+ Finetune	22.8083	0.72	0.253	25.2787 (+0.41)
+ Lightweight-tune	22.8027	0.56	0.253	28.4994 (+3.22)

Table 6. Ablation study on our proposed methods.

the score.

We compare the subjective quality of our proposed network with other SR networks in Fig. 6. The subjective results show sharp results compared to AsConvSR [20] and Bicubic++ [4] with less runtime, and comparable results to IMDN [27] and LRSRN [19] with superior performance.

4.3. Ablation Study

We conducted an ablation study to demonstrate the efficiency of our proposed methods. Sec. 4.1 shows the improvement in PSNR, runtime, MACs and score when the proposed methods are applied. Our baseline has 1 backbone and 32 channels. First, when applying down-sample, the PSNR is lower than the baseline, but the runtime is reduced by about 40% and the computation is reduced by 30%. When the cascade method of upsample is used, the amount of computation is reduced, but the runtime is slowed down and the PSNR is also degraded, which reduces the final score. On the other hand, when using our proposed upsample method, the computation amount is reduced by more than 70% compared to the direct upsample, and the PSNR performance is improved and the runtime is reduced. As a result, the score increases by 2.87 compared to the direct upsample and 4.56 compared to the cascade upsample. When our proposed RepConv is applied, the runtime and MACs are maintained and the PSNR increases by 0.029dB. Furthermore, we can see that fine tuning with the Distillation Scheme increases the PSNR by 0.024 dB compared to scratch. Finally, when we remove the bias in repconv to improve the runtime, the PSNR decreases by 0.006, which is not a significant difference, and the speedup is 0.16 ms faster. As a result, the score increases by 3.22 compared to finetune. Using our proposed methods, the runtime is over 65% faster and the computation is reduced by 85% compared to the baseline. The final score is an improvement of 10.20.

5. Conclusion and Future Work

In this paper, We proposed CASR, a low-computation and extremely fast real-time SR network that aims to upsample compressed images by a factor of 4. We used distil-

lation schemes, reparameterisation blocks, and proposed a novel upsampling method to speed up the runtime of the network and improve the performance. As a result, our network achieved a runtime of 0.56ms on the GPU, and achieved comparable performance to existing lightweight SR networks. We believe that the proposed network can perform the super resolution task in real-time on hardware-constrained devices used in the real industry.

In future work, we will validate that it can work on mobile or autonomous driving systems and other devices. Furthermore, we plan to extend our proposed network to perform other low level vision tasks such as denoising and deblurring in real-time, and to further improve our reparameterization approach.

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